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GENOTOXIC AND BIOCHEMICAL EFFECTS OF VERAPAMIL IN SWISS ALBINO MICE

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ABSTRACT

Verapamil, a calcium ion antagonist was evaluated for its effects on the genotoxic and biochemical activity in Swiss albino mice. Male mice were treated orally with different doses (25, 50 and 75 mg/kg, body weight/day) of verapamil for 7 days and sacrificed 24 hr after the last treatment. Femoral cells of mice were collected and studied for the frequency of micronuclei in polychromatic erythrocytes (PCE) and the ratio of PCE to normochromatic erythrocytes (NCE). Furthermore, proteins, DNA, RNA, malondialdehyde (MDA) and non-protein sulphydryl (NP-SH) were determined in hepatic cells. Veràpamil treatment failed to induce any significant changes in the frequency of micronuclei, and the concentrations of RNA, proteins and NP-SH. However, the ratio of PCE/NCE was significantly reduced. The hepatic levels of DNA were significantly decreased and the MDA concentrations were increased. The observed cytotoxic potentials of verapamil are attributed to the proxodant nature of verapamil in normal cells.

INTRODUCTION

Verapamil [2, 8-bis-(3,4-dimethoxyphenyl)-6-methyl-2-isopropyl-6- azaoctanitrile] a calcium ion antagonist is clinically used in the management of various diseases of the circulatory system, including hypertension, angina pectoris, cardiac arrythmias, cardiomyopathy, coronary and peripheral vasoconstriction. Furthermore, it is used as a modulator of p-glycoproteins (Fardel et al., 1994) to overcome drug reistance and potentiate the cytotoxicity of several antineoplastic drugs (Tidefelt et al., 1988; Dalton et al., 1989; Sweet et al., 1989; Huber et al., 1989; Hall et al., 1990). However, the verapamil-enhanced cytotoxicity has been found to accompany an increase in the chromosomal damage in the proliferating and cancerous cells (Sheid et al., 1984; Yalowich and Ross, 1984; Pradhan et al., 1987; Tidefelt et al., 1988; Weber et al., 1989; He et al., 1991). Verapamil-inuced potentiation of cytotoxicity and mutagenicity show its oxidant nature in proliferating and cancerous cells. Interestingly, in the same fast dividing cells verapamil has been demonstrated to be a potent antioxidant against stimulated oxygen radical injury (Mak et al., 1992; Foschi et al., 1993; Prabha et al., 1990; Stein et al., 1989).

A large number of papers have been published to establish the verapamil-induced increase inthe cytotoxicity and mutagenicity of antineoplastic drugs and its oxidant and antioxidant nature in the proliferating and cancerous cells. Nevertheless, despite the metabolic and physiological differences between dividing and the quiescent cells, there is paucity of literature on the effects of verapamil on the latter. Previous studies have demonstrated verapamil to induce teratogencity in chick embryos (Lee and Nagele, 1986) and potentiate the adriamycin-induced cardiac toxicity in rabbits (Stephens et al., 1987). Nishikawa et al. (1988) reported verapamil to potentiate the phenobarbital-induced cardiovascular teratogenicity in chick embryos. Furthermore it was found to deteriorate the ethanol-induced gastric ulcers in rats (Keo et al., 1986).

The present study was undertaken in view of; the reported literature on oxidant and antioxidant potentials of vreapamil in proliferating and cancerous cells; the metabolic and physiological differences between the normal and fast dividing cells; pharmacological importance of verapamil and paucity of its literature on the genotoxic and biochemical effects in normal femoral erythrocytes and hepatic cells of mice.

MATERIALS AND METHODS

Test chemicals:

Verapamil (Isoptin-Retard) was obtained from Knoll AG, Germany.

Animal stocks:

A total of 20 male mice (Swiss albino, bred at Experimental Animal Care Center, College of Pharmacy, King Saud University, Riyadh, Saudi Arabia) aged 6-8 weeks and weighing 25-30 g were used. The animals were randomly assigned to different control and treatment groups (Five mice in each group) that were maintained under tandard conditions of humidity (40-45%), temperature (23-25°C) and light (12 hr light/12 hr dark). The animals were fed with Purina chow diet and had free access to water.

Dose and treatment groups:

The doses selected (25-75 mg/kg, body weight) were based on the pharmacologically active doses reported in the literature (Brunner etal., 1991; Tsuruo, 1984). The different experimental groups of mice consisted of (1) Untreatedcontrol (Distilled water); (2) Verapamil, 25 mg/kg/da;(3) Verapamil, 50 mg/kg/day; (4) Verapamil, 75 mg/kg/day. Verapamil was administered by oral route for 7 days. The mice were sacrificed 24 hr after the last treatment.

Cytological procedure:

The micronucleus test procedure described by Schmid (1975) was used to determine clastogenicity in femoral cells. From the freshly killed animals, both the femora were removed in toto and freed from muscles. The femoral cells were aspirated in foetal calf serum as a fine suspension and centrifuged. After centrifugation, the cells were spread on slides and air dried. Coded slides were fixed in methanoland stained with May-Gruenwald solution followed by Gimsa staining. Polychromatic erythrocytes (PCE) were screened for micronuclei and normochromatic erythrocytes were recorded. The polychromatic/normochromatic cells ratio (P/N ratio) was calculated to analyse the bone marrow depression.

Biochemical procedure:

The livers from the same animals were quickly excised in liquid nitrogen and stored at-20°C until analysis.

Estimation of proteins an dnucleic acid:

Total protein was determined by the method of Lowryet al. (1951). The method of Bregman (1983) was used to determine the levels of nucleic acids. The tissues were homogenized in a glass tube with a motorized PTFE pestle and the homogenate was extracted in different concentrations of cold and hot trichloroacetic acid (TCA) and absolute ethanol. After the final extraction in TCA, incubation and centrifugation the supernatant was used to determine the levels of DNA and RNA. The levels of DNA were determined by treating the nucleic acid extract with diphenylamine reagent and reading the intensity of blue color at 600 nm. For quantification of RNA, the nucleic acid extract was treated with orcinol and the green color was read at 660 nm. RNA (B.D.H., UK) and DNA (Koch-light Labs., UK) were used to sketch the standard curves to determine the amounts of nucelic acids present.

Determination of malondialdehyde concentration:

The method described by Fong et al. (1973) was followed. The liver tissue was homogenized in TCA and the homogenate was suspended in thiobarbituric acid. After centrifugation, the optical density of the clear pink supernatant was read at 532 nm. Malondialdehyde bis (dimethyl acetal) (Fluka, Chemie Ag, Basel, Switzerland) was used as standard.

Quantification of nonprotein sulfhydryl levels:

The method described by Sedlak and Lindsay (1968) was used to determine the levels of NP-SH. The liver tissue was homogenized in ethylene diaminetetra acetic acid (EDTA) and mixed with TCA. The homogenate was centrifuged at 3000 x g. The supernatant was suspended in Trisbuffer and 5,5-5 dithiobis (2-nitrobenzoic acid) (DTNB) and the absorbance was read at 412 nm against reagent blank with no homogenate. Glutathione (Sigma, St. Louis, USA) was used as a standard.

RESULTS

The results obtained in the present study are summarized in Tables 1-4. Verapamil treatment failed to induce any significant changes in the frequency of micronucleicin polychromatic erythrocytes (PCE), whereas the ratio of PCE to normochromatic erythrocytes (NCE) was significantly reduced after treatment with verapamil at the higher doses (groups 3 and 4, Table 1).

The treatment with verapamil did not induce any significant changes in the hepatic levels of RNA and proteins (groups 2, 3, 4). Whereas the concentrations of DNA were significantly reduced at the higher dose ($P<0.05$, grup 4) as compared to the values obtained in the control (group 1, Table2).

The MDA levelsin the liver were found to significantly increase after the treatment with verapamil at the higher doses ($P<0.05$, groups 3 and 4) as compared to the values obtained in the control (grup 1, Table 3).

The administration of verapamil did not induce any significant changes in the hepatic concentrations of NP-SH ($P>0.05$) at any of the dose levels used(groups 2, 3, 4) as compared to the control (group 1, Table 4).

DISCUSSION

The results obtained in the present study showed that the treatment with verapamil failed to affect the incidence of micronucleated - PCE, indicating lack of any clastogenic activity. However, the ratio of PCE to NCE was significantly reduced demonstrating the cytotoxic potentials of verapamil. Although verapamil has been reported to synergistically enhance the mutagenic and cytotoxic efficiency of cytostatics such as bleomycin and peplomycin in cultured human lymphocytes (Scheid et al., 1984; Weber et al., 1989), mouse S-180 and HEP-2 cell lines (He et al., 1991), mitomycin in Chinese hamster ovary cells in in vitro (Scheid et al., 1991), 4'-demothylpipodophyllotoxin-9-(4.6-0-ethylidene-beta-D-glucopyranoside (VP-16) in L1210 cells in vitro (Yalowich and Ross, 1984), adriamycin in murine P388 leukemia, Ehrlich ascites carcinoma and Sarcoma-180, leukemic cells in humans in in vitro (Pradhan et al., 1987; Tidefelt et al., 1988), there is a paucity of literature on the clastogenic and cytotoxic potentials of verapamil in normal tissue in the absence of a stimulating factor.

Nevertheless, our data on the ratio of femoral erythrocytes observed in the present study clearly demonstrates the cytotoxic potentials of verapamil in normal cells. These data are supported by our results on the biochemical analysis, which showed verapamil to induce significant changes in the hepatic levels of DNA and MDA. The oxidant nature of verapamil, observed in the present study is not in corroboration with previous reports which confirms it santioxidant potentials in highly undifferentiated and rapidly proliferating cells in pro-oxidant conditions.

Many previous studies have established that increase in the rate of cell division is often associated with a decrased rate of lipid peroxidation (Slater, 1976; Morisaki et al., 1984; Cheeseman et al., 1986). Hence, highly significant reduction in lipid peroxidation has been demonstrated in proliferating and cancerous cells exposed to stimulating factors in pro-oxidant conditions, as compared to normal tissue (Grewela et al., 1975; Bartoli and Galeotti, 1979; Ahmed and Slater, 1981; Cheeseman et al., 1986; Prabha et al., 1990; Muller et al., 1991; Foschi et al., 1993). Inhibition of lipid peroxidation in such conditions has been described to be due to the increase in endogenous antioxidants, which is accompanied by reductions in cytochrome P-450 and polyunsaturated fatty acidds (Cheeseman et al., 1984). Furthermore, the reduction to cytochrome P450 in the proliferating cells may interfere with the biotransformation of verapamil and the activity of its reactive metabolites. Earlier studies have reported cytochrome P450 to be the major enzymes responsible for the metabolism of verapamil (Busse et al., 1995). Thus the discrepancy between the reports on the verapamil-induced free radical scavenging activity in proliferating cells and our results on its oxidant nature in normal cells appears to be due to thedeleterious effects of verapamil and its metabolites on endogenous antioxidants. Furthermore itis well known that the NADPH-cytochrome C is responsible for catalysing the NADPH + ADP + Iron-dependent lipid peroxidation. The antioxidant potentials of verapamil, observed in other studies on proliferating and cancerous cells may be related to the disruption of NADPH- cytochrome P-450 electron transport chain in cancer cells and these would most certainly produce less favurable conditions for initiation and/or propagation of lipid peroxidation in these preparations (Svinger et al., 1979).

The exact mechanism of verapamil-induced cytotoxicity and lipid peroxidation in the normal cells is not known. However it appears to be due to normal NADPH-cytochrome P-450 electron transport chain and the cytochrome P-450 which might have stimulated the genesis of free radicals. The other possibility is the biotransformation of verapamil resulting in exposure of the target tissue to the reactive metabolites. The present investigation suggests that verapamil is cytotoxic and induce lipid peroxidation in normal hepatic cells. Further studies are warranted to elucidate the exact mode of action of verapamil in normal erythrocytes and hepatic cells.

Table 1
Effect of verapamil on the frequency of micronuclei and the ratio of PCE and NCE in femoral cells of mice

Group No.	Treatment and dose (mg/kg/day)	Polychromatic Erythrocytes (PCE)	Micronucleated Polychromatic erythrocytes (%) Mean±S.E.)	Normochromatic Erythrocytes erythrocytes (NCE) screened	PCE/NE ratio (Mean±S.E.)
1.	Control (Distilled water, 0.3 ml/mouse)	5650	0.29±0.03	5650	1.08±0.04
2.	Verapamil (25.0)	5630	0.32±0.02	5360	1.05±0.03
3.	Verapamil (50.0)	5056	0.3±0.13	5506	0.91±0.04*
4.	Verapmil (75.0)	6000	0.34±0.06	6000	0.85±0.07*

Five mice were used in each group.
*P < 0.05 (Student's t-test).

Table 2
Effect of verapamil on hepatic levels of nucleic acids and proteins in Swiss albino mice

Group No.	Treatment and dose (mg/kg/day)	DNA (mg/100 mg) (Mean ± S.E.)	RNA (mg/100mg) (Mean ± S.E.)	Total proteins mg/100 mg (Mean ± S.E.)
1.	Control (Distilled water, 0.3 ml/mouse)	246.47 ± 9.00	743.43 ± 20.44	15.56 ± 0.33
2.	Verapamil (25.0)	239.56 ± 8.96	740.56 ± 22.30	15.60 ± 0.32
3.	Verapamil (50.0)	221.39 ± 8.76	748.78± 25.20	15.43 ± 0.15
4.	Verapamil (75.0)	211.43 ± 6.27*	679.97 ± 27.01	15.37 ± 0.24

#Five mice were used in each group.
*$P < 0.05$ (Student's test).

Table 3
Effect of verapamil on hepatic levels of malondialdehyde in Swiss albino mice

Group No.	Treatment and dose (mg/kg/day)	Malondialdehyde Concentration (n mol/g) Mean ± S.E.
1.	Control (Distilled water, 0.3 ml/mouse/day)	237.54 ± 9.69
2.	Verapamil (25.0)	240.65 ± 12.75
3.	Verapamil (50.0)	286.00 ± 17.11*
4.	Verapamil (75.0)	296.60 ± 14.97*

#Five mice were used in each group.
*$P < 0.05$ (Student's t-test).

Table 4
Effect of verapamil on hepatic levels of glutathione (NP-SH) in mice.

Group No.	Treatment and dose (mg/kg/day)	Malondialdehyde Concentration (m mol/g) Mean ± S.E.
1.	Control (Distilled water, 0.3 ml/mouse/day)	64.61 ± 2.97
2.	Verapamil (25.0)	60.72 ± 3.00
3.	Verapamil (50.0)	58.90 ± 2.70
4.	Verapamil (75.0)	59.75 ± 2.35

#Five mice were used in each group.
*$P < 0.05$ (Student's t-test).

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